

Free (left) h-adequate semigroups

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Ehresmann semigroups

Definition

A semigroup S equipped with an additional unary operation $^+$ is called *left Ehresmann* if it satisfies the identities

$$x^+x = x, \quad (x^+y^+)^+ = x^+y^+ = y^+x^+ \quad \text{and} \quad (xy)^+ = (xy^+)^+.$$

Examples: Any monoid with $m^+ = 1$, restriction semigroups, monoid of binary relations $\mathcal{B}_X$ with $\rho^+ = \{(x, x) \mid x \in \text{dom } \rho\}$, $\mathcal{P}(C)$ for a small category C with $A^+ = \{\text{Id}_{D(a)} \mid a \in A\}$...

Definition

A left Ehresmann semigroup $(S, ^+)$ is called *left adequate* if it additionally satisfies

$$x^2 = x \rightarrow x = x^+ \quad \text{and} \quad xz = yz \rightarrow xz^+ = yz^+.$$

Examples: Groups with $g^+ = 1$, inverse semigroups with $s^+ = ss^{-1}$, right cancellative monoids with $c^+ = 1$...

Free left Ehresmann monoids

Left Ehresmann monoids form a variety, so free objects exist in all ranks. Fix a set X .

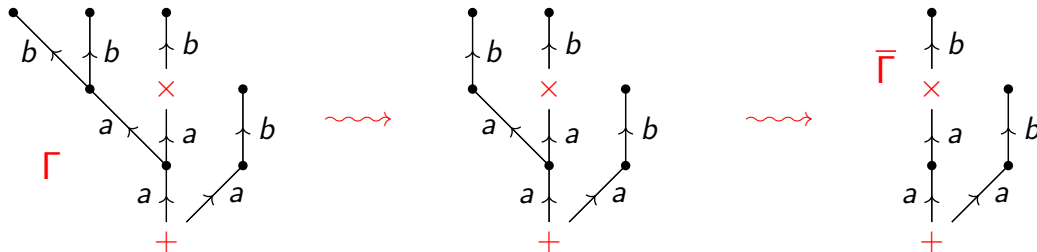
Definition

An X -tree is a directed tree, edge-labelled by X , with two vertices *start* and *end*, such that there is a path from the start to any given vertex.

Definition (Kambites, 2009)

The *pruning* of an X -tree Γ is a minimal image of Γ under idempotent endomorphisms fixing start and end.

Fact: The pruning of a tree Γ is unique up to isomorphism. We denote it $\bar{\Gamma}$.



Free left Ehresmann monoids II

Define a multiplication $\Gamma \Delta$ on X -trees as follows:

- 1 Glue Δ to Γ start-to-end.
- 2 Prune the result.

Define a unary operation Γ^+ on X -trees as follows:

- 1 Relocate the end of Γ to the start.
- 2 Prune the result.

Theorem (Kambites, 2009)

Fix a set X . Identify each $a \in X$ with the tree $+ \xrightarrow{a} \times$. The free left Ehresmann monoid on X is the set of all pruned X -trees with the operations above. Moreover, the free left adequate monoid on X is exactly the free left Ehresmann monoid on X .

Throughout, we will denote the free left Ehresmann monoid on X by $\text{FLE}(X)$.

Types of adequate monoids

Definition (Fountain, 1979)

A left adequate semigroup $(S, +)$ is called (*left*) *type A* if it additionally satisfies

$$xy^+ = (xy)^+x.$$

Type A semigroups are now called *ample*.

- Fountain, Gomes, Gould, 2009: Graphical description of free ample semigroups.
- Free objects coincide with free restriction semigroups.
- (Reasonably) well-studied by many people!

Definition (Fountain, 1979)

A left adequate semigroup $(S, +)$ is called (*left*) *type B* if it additionally satisfies

- 1 $(xz^+y)^+ = (xy)^+(xz)^+.$
- 2 If $e^2 = e = ea^+$, then $\exists x$ s.t. $e = (ax)^+.$

- Fountain constructs type B semigroups from general semilattices in certain ways.
- Otherwise these seem to have been mostly forgotten!
- Aird, H., 2026: The rank 1 free left Ehresmann monoid satisfies $(xz^+y)^+ = (xy)^+(xz)^+.$
- Aird, H., 2026: Free left Ehresmann monoids of rank ≥ 2 do not!

The h condition

Definition

A left Ehresmann semigroup $(S, +)$ is called *left h-Ehresmann* if it satisfies

$$(xz^+y)^+ = (xy)^+(xz)^+.$$

Examples: FLE_1 , inverse semigroups, left ample semigroups, left restriction semigroups, Schützenberger products of right cancellative semigroups (Fountain, Gomes, Gould, 1999).

Proposition (H., 2026)

Free **right** Ehresmann monoids in all ranks are left h-adequate.

And free objects...?

Theorem (Fountain, 1988)

The free left h-adequate semigroup on X is $(F_X * E_X) / \sim$ where F_X is the free semigroup on X , E_X is the semillattice of finite prefix codes over X , and $\sim$ is the $(2, 1)$ -congruence generated by $\{(a^+a, a) \mid a \in F_X * E_X\}$ (with details of $+$ on the free product omitted here).

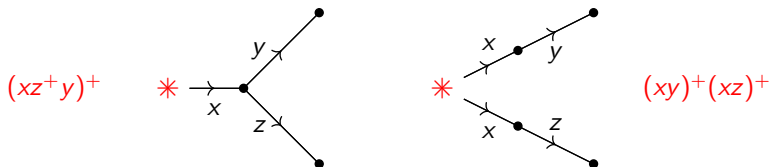
Fact: That's a lot of words. I like trees! Can we do it with trees?

The h condition on trees

Goal: Describe free left h-Ehresmann/h-adequate semigroups in terms of directed trees.

Why? Easier (in my opinion) visualisation of the word problem, elements, proof strategies etc.; give a description akin to those of its Ehresmann and ample neighbours; recover FLE_1 .

How? We already have a description for free left Ehresmann semigroups... so what does the h condition look like there?



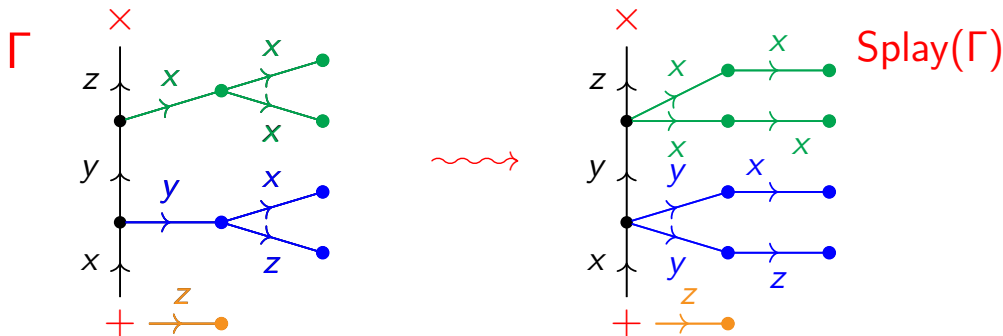
Our description will need to identify these trees...

Splaying

Definition (by example)

Given an X -tree Γ , define the tree $\text{Splay}(\Gamma)$ as follows...

- 1 Consider each *branch* of Γ ...
- 2 Transform each branch into the gluing of all of its paths to leaves.



The monoid $H(X)$

Define $H(X) = \{ \overline{\text{Splay}(\Gamma)} : \Gamma \text{ is an } X\text{-tree} \}$.

Define a multiplication $\Gamma \Delta$ as follows:

- 1 Glue Δ to Γ start-to-end.
- 2 Splay the result.
- 3 Prune the result.

Define a unary operation Γ^+ as follows:

- 1 Relocate the end of Γ to the start.
- 2 Splay the result.
- 3 Prune the result.

Theorem (H., 2026)

For non-empty X , the set $H(X)$ with its operations is the free left h-Ehresmann monoid on X .

Proof Idea. Construct a surjective morphism $\text{FLE}(X) \rightarrow H(X)$ and show the h condition. Then show freeness...! □

Properties

Proposition (H., 2026)

Free left h-Ehresmann monoids are left h-adequate.

Corollary

Free left h-Ehresmann and free left h-adequate monoids coincide in all ranks.

Gould, Johnson, 2027(!): Free left Ehresmann monoids are two-sided adequate.

Proposition (Fountain, 1988 + H., 2026)

Free left h-Ehresmann monoids are two-sided h-adequate.

Finitary conditions

Free left h-Ehresmann monoids enjoy various finitary properties.

Theorem (Gould, Johnson, 2027)

Free left Ehresmann monoids are...

- *Finitely left equated and finitely right equated.*
- *Weakly left coherent and weakly right coherent.*
- *Left ideal Howson and right ideal Howson.*

Theorem (H., 2026)

Free left h-Ehresmann monoids are...

- *Finitely left equated and finitely right equated.*
- *Weakly left coherent and weakly right coherent.*
- *Left ideal Howson and right ideal Howson.*

Proof idea. Construct an idempotent surjective morphism $\text{FLE}(X) \rightarrow H(X)$, then use a result of Dasar, Gould, Miller, 2025. □

Final thoughts

Gould, Johnson show that $\text{FLE}(X)$ is **not** left coherent for $\text{rank} \geq 2$.

The same is true for $H(X)$.

Gould, Hartmann, 2017: Free left **ample** monoids **are** right coherent...

Question

Are $\text{FLE}(X)$ and $H(X)$ right coherent?

Question

Is $\text{FLE}_1 = H_1$ left or right coherent?

Many thanks and happy birthday Vicky! ☺

D. Heath, [Free left h-Ehresmann semigroups](#). Preprint. [arXiv:2608.24657](#), 2026.

